Classification of Exoplanet Candidates: Assessing Kepler KOI Reliability via PCA and Machine Learning

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***Abstract*—This study presents a statistical quality assessment of Kepler Objects of Interest (KOIs) by classifying them into low- and high-quality categories based on their disposition scores. Using a dataset of 7,994 KOIs from the NASA Exoplanet Archive, we construct a binary classification task yielding 7,735 labeled examples (3,948 low quality, 3,787 high quality) after thresholding; this removes approximately 3% of ambiguous samples to ensure label purity. Importantly, this is a surrogate modeling task targeting extreme `koi_score` cases (pipeline behavior), not a direct prediction of physical truth (confirmed planet vs. false positive). We apply logarithmic and linear transformations to 12 astrophysical and observational features to address skewness and varying scales. Principal Component Analysis (PCA) is employed for dimensionality reduction; the first three principal components capture approximately 73.5% of the variance, while retaining four components explains approximately 81.8%. We benchmark multiple machine learning classifiers using an automated screening and tuning workflow, and select the strongest performers (LightGBM, Gradient Boosting, Random Forest, Extra Trees, and KNN) for evaluation on both the original 12-dimensional feature set and the PCA-reduced space. Our results demonstrate that while PCA provides a compact representation of the data structure, it results in a loss of predictive power compared to the full feature set. The best performance is achieved by a LightGBM classifier trained on the original features, attaining a test accuracy of 88.36% and an F1-score of 0.8834. These findings suggest that gradient boosting using catalog-level parameters can approximate Robovetter scoring behavior on `koi_score` extremes with $\approx 88\%$ test accuracy.**

***Index Terms*—Kepler Objects of Interest, disposition score, principal component analysis, classification, gradient boosting, exoplanet vetting, random forest**

I. Introduction

The NASA Kepler mission revolutionized exoplanetary science by monitoring over 150,000 stars to detect periodic dips in brightness caused by transiting planets [1]. This effort produced thousands of Kepler Objects of Interest (KOIs)—signals that resemble planetary transits but require rigorous vetting to distinguish genuine candidates from false positives, such as eclipsing binaries or instrumental artifacts [2], [3]. To automate this process, the Kepler team developed the Robovetter, which assigns a disposition score (`koi_score`) to each KOI. Ranging from 0 to 1, this score represents the confidence that a signal is a planet candidate [4]–[6].

While `koi_score` is a critical metric for statistical studies, its calculation relies on complex, computationally intensive pipelines involving simulated transit injections. This raises a practical question: can we effectively assess the quality of a KOI using only the standard catalog-level astrophysical and observational parameters? In this work, we explicitly treat this as a *surrogate modeling* problem: we model extreme `koi_score` cases to approximate Robovetter scoring behavior, rather than predicting confirmed planet truth. We aim to automatically classify KOIs into "low quality" (likely false positives) and "high quality" (likely candidates) categories using a set of 12 derived features, such as orbital period, transit depth, and signal-to-noise ratio.

This classification task presents several challenges. The feature space is high-dimensional and contains variables that span multiple orders of magnitude (e.g., planetary radius vs. orbital period). Furthermore, many features are physically correlated; for instance, transit depth is intrinsically linked to planetary radius and stellar size. These correlations suggest that the effective dimensionality of the problem may be lower than the number of features, motivating the use of dimensionality reduction techniques.

Our specific contributions are as follows:

- We construct a clean, balanced binary classification dataset from the continuous `koi_score` by applying strict thresholds to identify high-confidence examples.
- We perform a detailed exploratory data analysis and apply Principal Component Analysis (PCA) to 12 transformed features, demonstrating that a small set of orthogonal components captures the majority of the variance and reveals underlying physical relationships.
- We benchmark a diverse set of machine learning classifiers using an automated screening workflow, and select the top-performing models (LightGBM, Gradient Boosting, Random Forest, Extra Trees, and KNN) for final evaluation on both the original feature space and the PCA-reduced space.
- We provide an in-depth analysis centered on the best-performing model (LightGBM) and an interpretable comparator (Random Forest), including ROC analysis, confusion matrices, and feature importance, to interpret how astrophysical parameters drive automated KOI quality assessment.

II. Data Description

A. Dataset Overview

We utilize the Kepler Objects of Interest (KOI) cumulative table from the NASA Exoplanet Archive. The dataset initially comprises 7,994 KOIs. Our analysis focuses on 12 key astrophysical and observational features: orbital period (`koi_period`), transit duration (`koi_duration`), planetary radius (`koi_prad`), equilibrium temperature (`koi_teq`), insolation flux (`koi_insol`), stellar radius (`koi_srad`), transit depth (`koi_depth`), model signal-to-noise ratio (`koi_model_snr`), impact parameter (`koi_impact`), stellar effective temperature (`koi_steff`), stellar surface gravity (`koi_slogg`), and Kepler magnitude (`koi_kepmag`).

The target variable is the continuous `koi_score`. To formulate a binary classification problem, we threshold this score to separate clear false positives from likely candidates. We define "Low Quality" (class 0) as samples with `koi_score` < 0.3 and "High Quality" (class 1) as samples with `koi_score` > 0.7, discarding ambiguous samples in between. This process removes 259 ambiguous samples ($\approx 3.2\%$ of the original data), yielding a final labeled dataset of 7,735 samples (3,948 class 0, 3,787 class 1). This strict thresholding ensures that our models are trained on high-confidence labels, minimizing noise in the ground truth.

B. Data Cleaning and Missing-Value Handling

Prior to modeling, we perform basic data cleaning to ensure a consistent feature table. We coerce the selected columns to numeric types when appropriate, drop rows with missing `koi_score`, and filter the target to the valid range $[0, 1]$. Because the downstream pipeline requires a complete 12-feature vector, we drop any rows missing one or more of the selected feature columns. We do not apply explicit outlier removal beyond the stabilizing log/log1p transformations described below.

Figure 1 illustrates the distribution of the original `koi_score`, and Table I summarizes the key statistics.

TABLE I
SUMMARY STATISTICS OF `KOI_SCORE`

Statistic	Value
Count	7994
Mean	0.484
Std Dev	0.477
Min	0.000
25%	0.000
50%	0.371
75%	0.999
Max	1.000

C. Feature Transformations

Many exoplanet features follow power-law distributions or span several orders of magnitude. To improve the performance of linear models and PCA, we apply the following transformations:

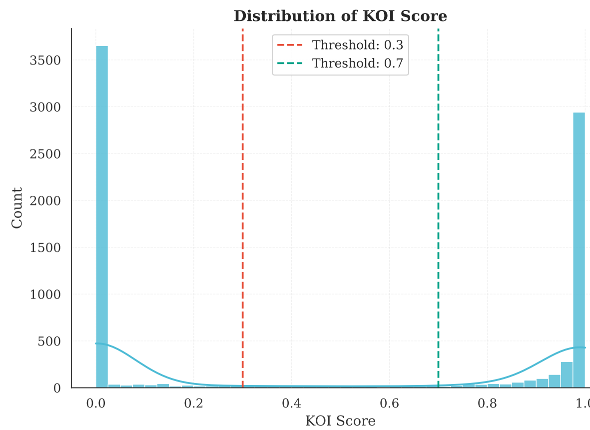

Fig. 1. Histogram of `koi_score`. The bimodal nature facilitates binary labeling by thresholding.

- **Log_{10}:** Applied to `koi_period`, `koi_duration`, `koi_prad`, `koi_teq`, `koi_insol`, `koi_srad`.
- **$\mathrm{Log}_{10}(1 + x)$:** Applied to `koi_depth` and `koi_model_snr` to handle zero-bounded values with wide dynamic ranges.
- **No Transformation:** Applied to `koi_impact`, `koi_steff`, `koi_slogg`, `koi_kepmag`.

Figure 2 shows the distributions of these features after transformation.

D. Correlation Analysis

The transformed features exhibit significant correlations, as shown in Figure 3. For example, planetary radius (`koi_prad`) is strongly correlated with transit depth (`koi_depth`) and signal-to-noise ratio (`koi_model_snr`). Similarly, stellar parameters like radius and surface gravity show expected physical correlations. These redundancies support the use of PCA to reduce dimensionality while preserving information.

III. Methodology

A. Problem Formulation

We formally define the task as a binary classification problem. Let $\mathbf{x} \in \mathbb{R}^{12}$ represent the vector of transformed astrophysical features for a given KOI, and let $y \in \{0, 1\}$ be

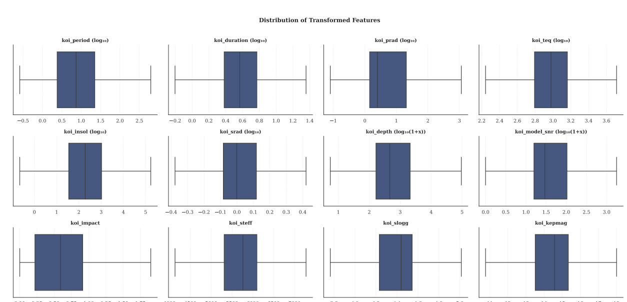

Fig. 2. Boxplots of the 12 features after logarithmic and linear transformations.

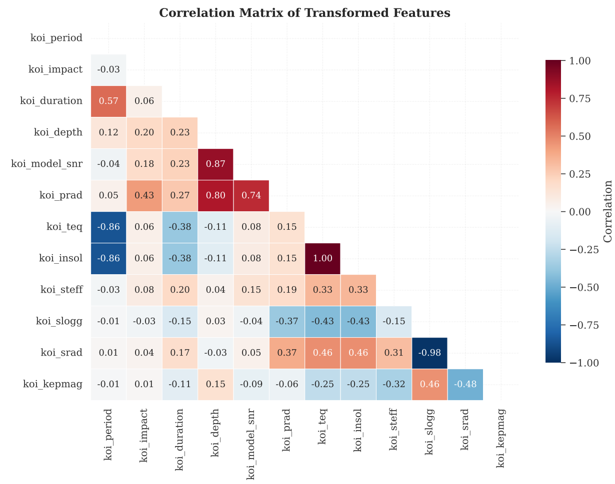

Fig. 3. Correlation matrix of the 12 transformed features.

the corresponding quality label, where 0 denotes a low-quality candidate (likely false positive) and 1 denotes a high-quality candidate. Our goal is to learn a mapping $f : \mathbb{R}^{12} \rightarrow \{0, 1\}$ that minimizes the classification error on unseen data. The labels are derived from the continuous variable $\texttt{koi\_score} \in [0, 1]$ using a thresholding function:

$$y_i = \begin{cases} 1 & \text{if } \texttt{koi\_score}_i > 0.7 \\ 0 & \text{if } \texttt{koi\_score}_i < 0.3 \\ \text{undefined} & \text{otherwise} \end{cases} \tag{1}$$

Samples falling in the ambiguous range $[0.3, 0.7]$ are discarded to ensure high label quality. While this reduces the dataset size slightly, it ensures that the models are trained on confident ground-truth labels, avoiding noise from uncertain dispositions.

B. Principal Component Analysis

Principal Component Analysis (PCA) is a linear dimensionality reduction technique that projects data onto a lower-dimensional subspace defined by the directions of maximum variance [7]. Given the standardized feature matrix $\mathbf{X} \in \mathbb{R}^{N \times D}$, where N is the number of samples and $D = 12$ features, PCA computes the covariance matrix $\mathbf{C}$:

$$\mathbf{C} = \frac{1}{N-1}\mathbf{X}^T\mathbf{X} \tag{2}$$

We then compute the eigenvalues λ_j and eigenvectors $\mathbf{v}_j$ of $\mathbf{C}$. The eigenvectors (principal components) define a new orthogonal coordinate system. We project the data onto the top k components corresponding to the largest eigenvalues. We select $k = 4$ to satisfy a cumulative explained variance threshold of $\geq 80\%$, balancing information retention with dimensionality reduction.

C. Classification Models

We employ several classifier families and select the top performers from an initial screening stage. In the final experiments we focus on gradient boosting and ensemble/tree-based methods, complemented by a distance-based baseline (KNN):

1) Random Forest (RF): An ensemble learning method that constructs a multitude of decision trees during training [11]. It handles nonlinear interactions between features effectively and is robust to outliers. We use it to assess the importance of original features and to capture complex boundaries that linear models might miss. We utilize the Gini impurity criterion for splitting nodes and evaluate feature importance by measuring the mean decrease in impurity.

2) Light Gradient Boosting Machine (LightGBM): A gradient boosting framework that uses tree-based learning algorithms [10]. Unlike traditional boosting, LightGBM grows trees leaf-wise rather than level-wise, often resulting in lower loss and faster training. It was included in the final analysis due to its superior performance in preliminary benchmarks.

3) Gradient Boosting Classifier (GBC): A classical gradient boosting ensemble that builds an additive model of shallow decision trees in a stage-wise fashion. It provides a strong nonlinear baseline and is competitive on tabular data.

4) Extra Trees (ET): An ensemble of randomized decision trees similar to Random Forest, but with additional randomization in split selection. This can reduce variance and often performs well on noisy tabular features.

5) K-Nearest Neighbors (KNN): A non-parametric, distance-based classifier that predicts labels based on the majority class among the k nearest points in feature space. KNN benefits from standardized inputs and serves as a simple baseline for local neighborhood structure.

D. Evaluation Metrics

Because the dataset is roughly balanced, we use Accuracy as a primary metric. However, to account for potential costs of false positives versus false negatives in exoplanet vetting, we also report:

- **Precision:** $TP/(TP + FP)$. The fraction of claimed candidates that are real.
- **Recall (Sensitivity):** $TP/(TP + FN)$. The fraction of real candidates successfully recovered.
- **F1-Score:** $2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$. The harmonic mean of precision and recall.
- **AUC (Area Under the ROC Curve):** A threshold-independent measure of discriminative ability.

We also report Precision, Recall, and F1-Score, which together capture the trade-off between correctly identifying high-quality candidates and avoiding false alarms [12].

IV. Experimental Setup

A. Preprocessing Pipeline

The data preprocessing and modeling pipeline is implemented in Python using `scikit-learn` [8] and `PyCaret` [9]. The dataset is split into a training set (80%) and a held-out test set (20%) using stratified sampling to maintain class balance.

1) **Standardization:** All continuous features are scaled to have zero mean and unit variance. The scaler is fit only

on the training data and then applied to the test data to prevent data leakage.

2) **PCA Projection:** The PCA transformation is learned from the standardized training data. We retain the first 4 principal components, which explain over 80% of the total variance.

B. Cross-Validation and Hyperparameter Tuning

We employ 10-fold cross-validation on the training set to evaluate model stability and tune hyperparameters.

- **LightGBM:** We tune core boosting and tree-structure parameters (e.g., number of estimators, learning rate, and leaf constraints) to optimize F1-score.
- **Gradient Boosting (GBC):** We tune the number of boosting stages and tree depth-related parameters to balance bias and variance.
- **Random Forest (RF):** We tune the number of trees (n_estimators), maximum depth, and minimum samples per split to balance bias and variance.
- **Extra Trees (ET):** Similar to RF, we tune ensemble size and depth-related parameters; ET increases randomness by using random split thresholds.
- **K-Nearest Neighbors (KNN):** We tune the number of neighbors and distance weighting to optimize F1-score.

For completeness, the tuning process searches over model-specific hyperparameters (e.g., learning rate and tree complexity for boosting models, tree depth and sampling parameters for ensembles, and neighborhood size for KNN) and selects configurations that maximize F1-score on cross-validation folds.

C. Model Selection Protocol

Our experimental protocol consists of two phases:

1) **Broad Screening:** We initially compare over 10 different classifiers (including Gradient Boosting, SVM, KNN, etc.) using default hyperparameters to identify top performers on the original 12-dimensional feature set.
2) **Focused Analysis:** We select the top-performing models from the screening stage (LightGBM, Gradient Boosting, Random Forest, Extra Trees, and KNN). We tune each model using cross-validation (optimizing F1-score) and then evaluate the tuned models on both the original features and the 4-component PCA space using the held-out test set. For interpretability, we emphasize Random Forest feature importance while using LightGBM as the strongest overall performer.

V. Results

A. Principal Component Analysis

The eigenvalue decomposition of the standardized feature covariance matrix reveals that the effective dimensionality of the data is significantly lower than 12. As shown in Table II, the first three principal components (PCs) explain approximately 73.5% of the total variance, and the first four explain approximately 81.8%. This justifies our choice of $k = 4$ for the reduced dimensionality experiments, as we retain over 80% of the signal while reducing the feature count by two-thirds.

TABLE II
PCA Eigenvalues and Explained Variance

Component	Eigenvalue	Explained %	Cumulative %
PC1	3.735	31.1	31.1
PC2	2.995	25.0	56.1
PC3	2.083	17.4	73.5
PC4	1.003	8.4	81.8
PC5	0.923	7.7	89.5

The Scree Plot (Figure 4) and Pareto Plot (Figure 5) confirm this trend, with a distinct "elbow" appearing after the third component.

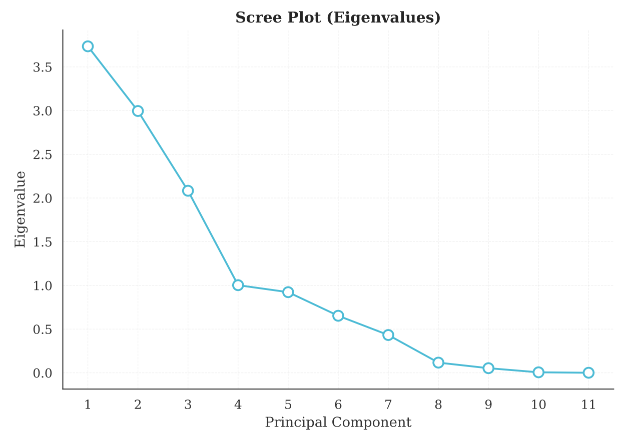

Fig. 4. Scree Plot showing eigenvalues for each principal component.

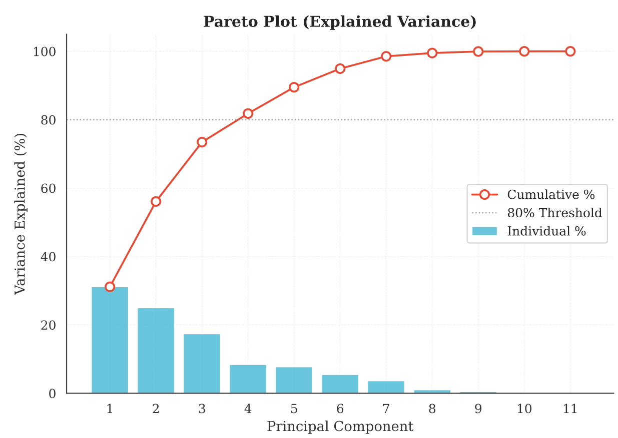

Fig. 5. Pareto plot of explained variance. The dashed line marks the 80% threshold, achieved by 4 PCs.

Analysis of the factor loadings (Figure 6) allows us to interpret the physical meaning of these components:

- **PC1 (31.1%):** Dominated by stellar environment variables (`koi_teq`, `koi_insol`, `koi_srad`).
- **PC2 (25.0%):** Heavily weighted by signal strength features (`koi_depth`, `koi_model_snr`) and planetary radius (`koi_prad`).

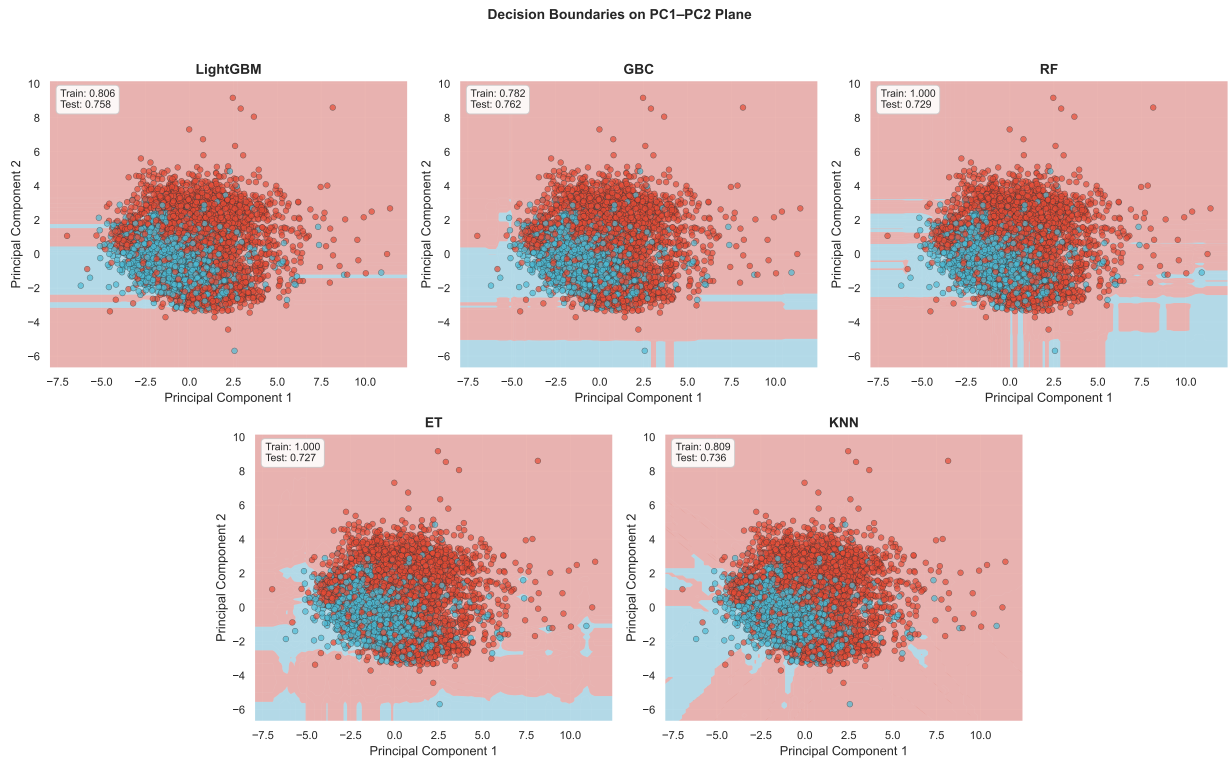

Fig. 7. Decision boundaries of LightGBM, RF, and others in the PC1-PC2 latent space.

- **PC3 (17.4%):** A mix of orbital characteristics and stellar properties (`koi_period`, `koi_slogg`).

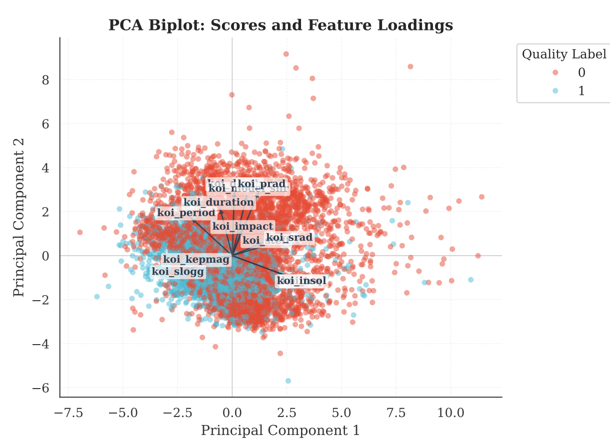

Fig. 6. PCA Biplot showing the first two principal components. Vectors indicate the direction and strength of feature contributions.

B. Model Benchmarking

We first benchmarked multiple classifiers using 10-fold cross-validation. Table III summarizes the results on the original 12-dimensional feature set. Among the candidate models benchmarked using PyCaret, LightGBM achieved the highest performance, closely followed by ensemble methods like Extra Trees and Random Forest.

TABLE III
TOP MODEL PERFORMANCE (ORIGINAL FEATURES)

Model	Accuracy	AUC	Recall	Prec.	F1
LightGBM	0.8751	0.9417	0.8939	0.8576	0.8752
Extra Trees	0.8686	0.9364	0.8873	0.8516	0.8688
Gradient Boosting	0.8668	0.9338	0.8949	0.8436	0.8681
Random Forest	0.8684	0.9358	0.8798	0.8565	0.8677
K Neighbors	0.8601	0.9147	0.8902	0.8357	0.8619

When trained on the 4-PC subset, performance generally decreased. Table IV summarizes the corresponding 10-fold cross-validation results on the PCA feature set. In this setting, LightGBM accuracy drops to $\approx 80.5\%$ and Random Forest to $\approx 80.3\%$, indicating that while the first 4 PCs capture most variance, the remaining 20% contains fine-grained information useful for classification.

TABLE IV
TOP MODEL PERFORMANCE (PCA FEATURES, 4 PCS)

Model	Accuracy	AUC	Recall	Prec.	F1
LightGBM	0.8047	0.8774	0.8364	0.7810	0.8075
Gradient Boosting	0.8010	0.8694	0.8369	0.7756	0.8047
Random Forest	0.8026	0.8750	0.8157	0.7897	0.8019
Extra Trees	0.7994	0.8740	0.8157	0.7845	0.7994
K Neighbors	0.7945	0.8578	0.8076	0.7810	0.7938

C. Test Set Performance

Table V presents the final evaluation of our selected models on the held-out test set (20% split; 1,547 examples), reported for both the original 12-dimensional features and the PCA-reduced representation (4 PCs). LightGBM trained on original features achieved the highest overall performance, with an F1-score of 0.8834.

TABLE V
FINAL TEST SET PERFORMANCE

Model	Features	Accuracy	F1	AUC
LightGBM	**Original (12-D)**	**0.8836**	**0.8834**	**0.9413**
LightGBM	PCA (4 PCs)	0.7957	0.8000	0.8621
Gradient Boosting	Original (12-D)	0.8830	0.8830	0.9422
Gradient Boosting	PCA (4 PCs)	0.7983	0.8033	0.8655
Random Forest	Original (12-D)	0.8707	0.8701	0.9387
Random Forest	PCA (4 PCs)	0.7951	0.8037	0.8634
Extra Trees	Original (12-D)	0.8694	0.8700	0.9361
Extra Trees	PCA (4 PCs)	0.7835	0.7926	0.8524
KNN	Original (12-D)	0.8746	0.8769	0.9237
KNN	PCA (4 PCs)	0.8093	0.8155	0.8745

D. Model Diagnostics: Confusion Matrices and ROC

To better understand the error modes of our models, we examine the confusion matrices for the selected classifiers on the original feature set (Figure 8).

- **LightGBM** shows balanced performance, effectively minimizing both false positives and false negatives.
- **Random Forest** and **Extra Trees** show similar patterns, confirming the robustness of ensemble tree-based methods for this dataset.

The ROC curves in Figure 9 illustrate the discriminative ability of the models. The curve for LightGBM extends well towards the top-left corner (AUC > 0.94), while Random Forest is close behind with an AUC slightly below 0.94. This confirms that the models are highly effective at ranking candidates, capable of maintaining high true positive rates while keeping false positive rates low.

E. Decision Boundaries and Feature Importance

The decision boundaries in the PC1-PC2 plane (Figure 7) visualize how the models separate the classes in the latent space. LightGBM and Random Forest demonstrate the ability to form complex, non-linear decision regions that capture the dense clusters of high-quality candidates.

Feature importance analysis from the Random Forest model (Figure 10) confirms that planetary radius (`koi_prad`) is the most critical predictor (importance ≈ 0.20), followed by transit depth (`koi_depth`) and period (`koi_period`). This aligns with physical intuition: larger planets with deeper, clearer transits are easier to confirm.

VI. DISCUSSION

A. Dimensionality Reduction vs. Predictive Power

Our analysis demonstrates a clear trade-off between model simplicity and predictive accuracy. While the first four principal components capture nearly 82% of the variance in the feature space, models trained on this reduced set consistently underperformed those trained on the full 12-dimensional dataset (e.g., LightGBM F1 dropped from 0.88 to 0.80). This suggests that the ”minor” components (PC5-PC12), which account for the remaining 18% of variance, contain subtle but discriminative information essential for distinguishing difficult borderline cases. PCA, being an unsupervised technique, maximizes variance rather than class separability; thus, the directions of maximum variation are not necessarily the directions of maximum classification margin.

B. Model Performance and Complexity

The comparison between classifiers highlights the effectiveness of gradient boosting for this task. LightGBM outperformed other models, achieving an F1-score of 0.8834 on the test set. This superior performance can be attributed to its ability to handle non-linear interactions and complex feature distributions more effectively than standard ensemble methods or simpler baselines. The inclusion of LightGBM in our final analysis addresses the need for a high-performance surrogate model, while Random Forest remains valuable for interpretability through feature importance analysis.

C. Astrophysical Interpretation

The feature importance analysis reinforces standard exoplanet detection theory. Planetary radius (`koi_prad`) and transit depth (`koi_depth`) emerged as the most influential predictors. This is intuitive: larger planets block more light, creating deeper and more significant signals that are less likely to be confused with noise or instrumental artifacts. The prominence of signal-to-noise ratio (`koi_model_snr`) further supports this, as high-SNR events are inherently more reliable.

D. Limitations

A limitation of our study is the exclusion of ambiguous candidates with scores between 0.3 and 0.7. While this ensures that our models are trained on high-confidence labels, it means the model is not tested on the ”hardest” cases where the Robovetter itself is uncertain. Future work should focus on these edge cases, potentially using regression to predict the exact `koi_score` rather than a binary classification.

More fundamentally, our model is a *surrogate* for the Kepler vetting pipeline: it predicts a thresholded version of the Robovetter disposition score rather than physical truth (i.e., confirmed planet vs. astrophysical false positive). As a result, strong performance should be interpreted as agreement with the pipeline's scoring behavior on the selected subset, not as definitive astrophysical validation.

There is also a risk of *label leakage* in the broad sense that some catalog-level features may overlap with, or be strongly correlated with, internal tests and derived quantities that influence `koi_score`. While we do not have full visibility into the score's internal computation, this possibility limits causal interpretation of feature importance and may inflate apparent predictive performance.

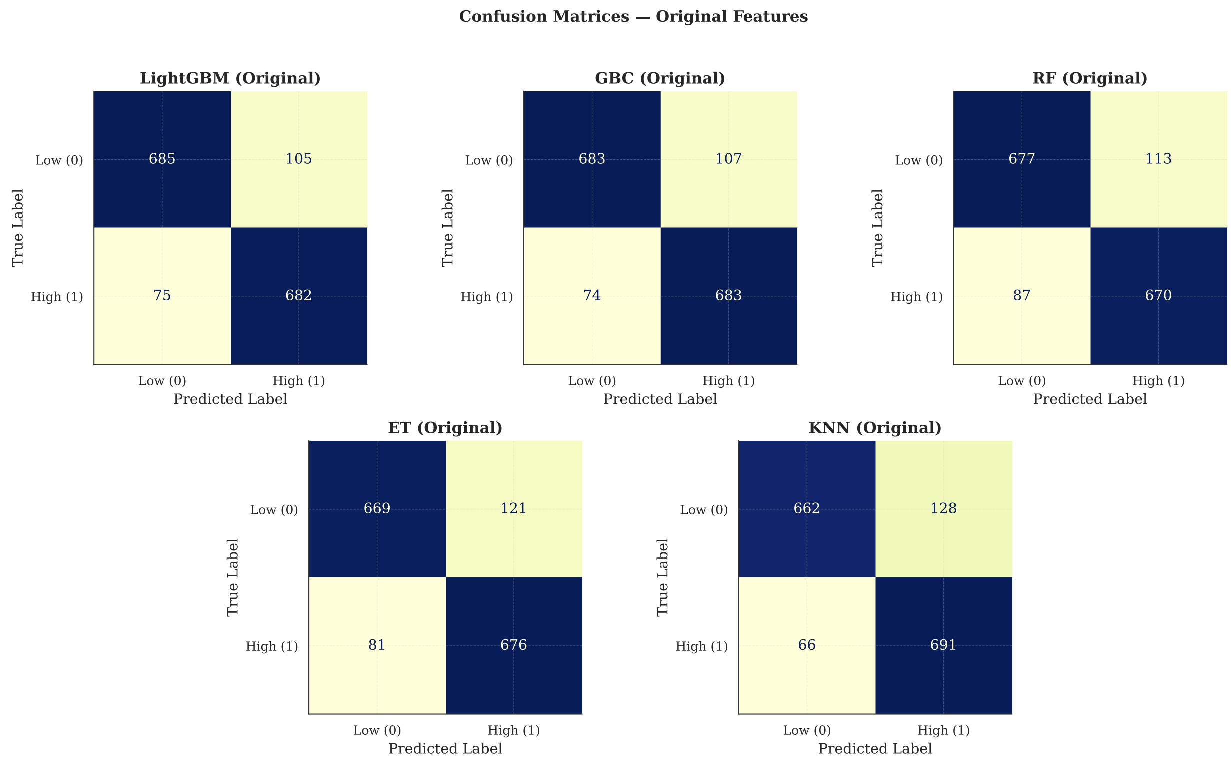

Fig. 8. Confusion Matrices for top models on the test set (Original Features).

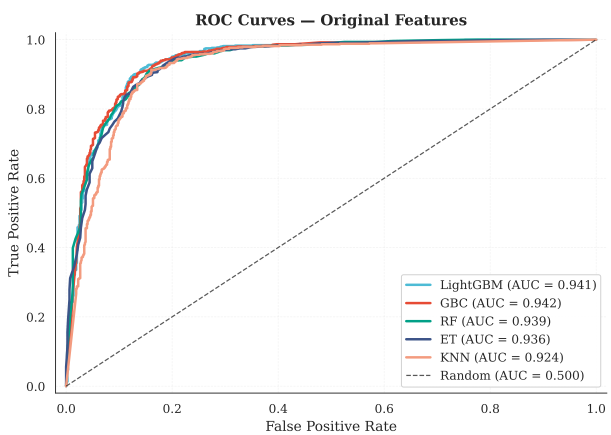

Fig. 9. ROC Curves for the models trained on original features.

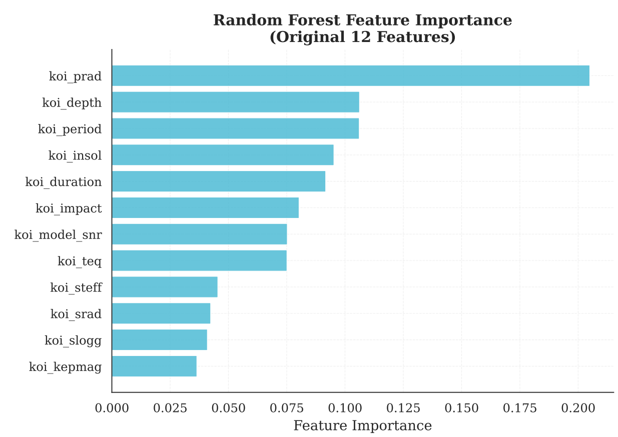

Fig. 10. Random Forest feature importance (Original Features).

In addition, generalization remains uncertain. The model is trained on a specific Kepler catalog snapshot and feature definitions; its performance may degrade under dataset shift, such as later catalog updates, changes in data processing, or application to other missions (e.g., K2 or TESS) with different noise properties and observing strategies.

Finally, the observed performance differences among the top models are relatively small (e.g., LightGBM F1 ≈ 0.883 vs. tree ensembles near ≈ 0.87 on one held-out split). Without uncertainty estimates (e.g., repeated train/test splits, confidence intervals, or paired statistical tests), it is unclear whether LightGBM is meaningfully better or only marginally ahead due to sampling variability. Relatedly, we do not evaluate probability calibration or decision-threshold selection; for a reliability-focused task, calibration diagnostics (e.g., reliability diagrams or Brier score) and explicit precision–recall trade-off curves would provide more actionable guidance than a single fixed threshold.

VII. Conclusion and Future Work

A. Conclusion

This study aimed to determine whether the reliability of Kepler Objects of Interest, as measured by the Robovetter's `koi_score`, could be predicted using a lightweight machine learning approach based on standard catalog attributes. We successfully constructed a balanced binary classification dataset and demonstrated that:

- Principal Component Analysis is effective for exploring the physical structure of the KOI data, with the first three components explaining $\approx 73.5\%$ of the variance.
- While PCA simplifies the feature space, it discards information critical for high-accuracy classification. Models trained on the original 12-dimensional feature set consistently outperformed those trained on the 4-dimensional PCA projection.
- Gradient boosting classifiers, specifically LightGBM, achieved strong performance with test accuracies of $\approx 88.4\%$ and F1-scores of ≈ 0.88, outperforming simpler baselines such as KNN.

We conclude that a data-driven surrogate model, particularly LightGBM, can approximate Robovetter scoring behavior on `koi_score` extremes using only high-level catalog parameters, achieving $\approx 88\%$ test accuracy on a held-out split.

B. Future Work

To further improve the robustness and utility of this approach, we propose the following directions:

- **Deep Learning on Light Curves:** Extending beyond catalog-level summary statistics, deep learning models (e.g., 1D-CNNs or Recurrent Neural Networks) could be applied directly to raw light curves and centroid time-series to better approximate the full Kepler vetting pipeline.
- **Ensemble Stacking:** Combining predictions from heterogeneous models (e.g., LightGBM, Random Forest, and KNN) via a meta-learner could potentially push classification accuracy beyond the 90% threshold.
- **Cost-Sensitive Learning:** Implement cost-sensitive training to explicitly penalize false positives (classifying a false alarm as a candidate) or false negatives (missing a real planet), depending on the specific scientific goal (purity vs. completeness).
- **Feature Engineering:** Incorporate time-series features extracted directly from light curves or centroid time-series data to better mimic the detailed tests performed by the Robovetter.

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